

Independent Verification of the Blitz T-DCA ETHUSD Perpetual Backtest

One published ETH preset | four cost scenarios | 2021-01-01 00:00 UTC to 2026-08-25 00:00 UTC (exclusive)

Author: Ranveer Verma (github.com/ranveer9) Date: 2026-09-30 Version: v1.0.0 Repository: <https://github.com/ranveer9/blitz-eth-tdca-verification>

Commissioned by: Blitz Trading. The work was commissioned and paid for by Blitz Trading; the method, results and conclusions are the author's own. This is a historical verification of a backtest. It is not a guarantee of live trading performance.

Edition: PUBLIC EDITION - exact settings, arrays and cycle records omitted

Summary

Phase A - reproduction under the original assumptions: reproduced. An independent implementation of the written specification and the client's written clarifications, run on 2,062 checksum-verified Binance daily ETHUSD aggTrades files (2,963,456,806 trades, identical to the reference count), reproduces the reference bit-for-bit in all four cost scenarios for every cycle count, reset, fee, slippage, drawdown, liquidation-room, MAE, utilisation, ladder-depth and completed-cycle holding statistic (all exact in all four scenarios), and for the baseline total turnover and the baseline and funding annual turnover. The baseline realised wallet is identical to the last digit (1,947,269.7970803205 USD).

Two residual differences are explained, not adjusted. (1) In the three funding scenarios the realised wallet differs by +7.29 / +6.66 / +10.91 USD; each gap equals the funding-paid difference to ten decimals, so it comes entirely from the recovered funding-mark inputs (the original funding files were deleted by the client). (2) The published exit-net terminal equity was valued at the close of the 2026-08-25 00:00 minute, one minute inside the sealed period, because the reference runner reads mark candles up to one minute past the last trade. The client confirmed this boundary inconsistency. This report values at the 2026-08-24 23:59 close, the last minute of the stated window; the corrected baseline exit-net terminal equity is 1,946,324.41 USD (published 1,946,288.58).

Phase B - adequacy of the assumptions. Reproducing the model does not validate it as a description of exchange execution. Risk uses 1-minute mark closes applied from the start of each minute (information not available at that time) with interpolated marks and a flat maintenance rate; fills are assumed complete when a trade reaches the order price. Quantified sensitivities on baseline costs: F1 one-tick trade-through: exit-net 1,970,210.55 USD (+1.23%); F2 volume at/through $\geq 1\times$ size: exit-net 1,135,467.07 USD (-41.66%); F3 volume at/through $\geq 2\times$ size: exit-net 1,060,026.55 USD (-45.54%); L1 1-second latency: exit-net 797,559.09 USD (-59.02%). No run was liquidated under the reference proxy; that does not demonstrate exchange-level survival.

Publication boundary. At the client's instruction this public edition omits the exact ladder distances, allocations, configuration values and all cycle-level records. The accompanying public code is generic and reads every setting from the private configuration; this public package alone cannot reproduce the results. The omission does not limit any material finding.

1. Scope, inputs and data integrity

Frozen scope: Binance ETHUSD USD-M perpetual, long-only T-DCA, one published preset, four cost scenarios (baseline; historical funding; funding + 1 bp TP slippage; funding + 2 bp TP slippage). Window 2021-01-01 00:00:00 UTC inclusive to 2026-08-25 00:00:00 UTC exclusive. The sealed out-of-sample period from 2026-08-25 was not downloaded, inspected or used.

Handover: all 17 client files were verified against the client's SHA256SUMS before every run. Market data: 2,062 daily aggTrades archives were downloaded from data.binance.vision and verified against their official .CHECKSUM files (download log: OK 2,062). The run processed 2,963,456,806 trades; the reference reports 2,963,456,806 (identical).

Recovered inputs (client-supplied). The client deleted the original large market files; funding and 1-minute mark data were recovered from official Binance sources. Funding: 6,186 events; 3,086 marks came directly from the funding API and 3,100 empty historical marks were backfilled with the official 1-minute mark OPEN. Marks: 2,969,280 consecutive minutes; the original archives lacked 4,333 minutes, of which 4,320 were filled with exact USD-M mark API observations and 13 (about 0.000438%) are semantic proxies (5 USD-M index, 8 COIN-M mark), not exact ETHUSD USD-M marks. These inputs carry current recovery checksums, not evidence of byte identity with the deleted originals.

2. Method

Independence. The engine was written in Python 3.12 (compiled with Numba) from the written specification and the client's written answers to ten clarification questions. The client's engine source was not provided or consulted. The client supplied the exact metric definitions and, after the first full run, one unstated rule (see 2.2) and a one-day cycle list used only to locate a difference.

2.1 Event model reproduced

Per daily file, minutes from the first trade's minute to one minute past the last trade are processed. At each minute start risk is updated with that minute's mark CLOSE (even while flat), then funding events before the minute end are applied with each record's own mark (risk checked before and after payment), then the minute's trades are processed with that close as their mark. Flat: the full executable ladder is previewed, the first entry fills at the trade price plus a fixed slippage rounded up to the tick, sized at the observed price, and the next trade is scanned. Open: the scan finds the first trade at or through the next ladder level or at or above the take-profit price; the segment from the previous trade fills all crossed levels in sequence

with marks interpolated by price fraction and liquidation checked on the path, or executes the take-profit. Wallet resets upward after a growth threshold. Terminal wealth is exit-net equity valued at the final mark.

2.2 One rule found during verification

The first full run reproduced the trade count but not the totals (baseline realised wallet 0.72% lower, 122 fewer cycles). A diagnostic of 20 rule variants against the client's exact 2021 annual figures ruled out all pricing and boundary hypotheses. The client then stated a rule absent from the specification: triggers compare the float64 tick price ($k \times 0.01$) with the parsed trade price with no tolerance. When $k \times 0.01$ lands one ulp above the decimal price, a trade exactly at that decimal price does not trigger. With that rule the client's 2021-01-01 cycle list was reproduced exactly (15 cycles, wallet identical to 17 digits) and the full run then matched. This is part of the reference behaviour; its execution meaning is assessed in 4.2.

Comparison tolerances: counts, timestamps and booleans exact; monetary and percentage values MATCH if the absolute difference is at most $1e-6$ or the relative difference at most $1e-12$.

3. Phase A - reproduction under the original assumptions

3.1 Scenario comparison with the client reference

Field	Baseline	Funding	Funding + 1 bp	Funding + 2 bp
ticks_processed	MATCH	MATCH	MATCH	MATCH
final_wallet	MATCH	EXPLAINED (funding data)	EXPLAINED (funding data)	EXPLAINED (funding data)
published_final_wallet_usd(exit-net)	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)
exit_net_return_pct	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)
annualized_exit_net_return_pct	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)
cycles_started	MATCH	MATCH	MATCH	MATCH
cycles_completed_tp	MATCH	MATCH	MATCH	MATCH
reset_count	MATCH	MATCH	MATCH	MATCH
fees_total	MATCH	MATCH	MATCH	MATCH
funding_paid_signed	MATCH	EXPLAINED (funding data)	EXPLAINED (funding data)	EXPLAINED (funding data)
slippage_cost	MATCH	MATCH	MATCH	MATCH
max_drawdown_pct	MATCH	MATCH	MATCH	MATCH
min_liquidation_room_pct	MATCH	MATCH	MATCH	MATCH
max_step	MATCH	MATCH	MATCH	MATCH
max_hold_days	MATCH	MATCH	MATCH	MATCH
hold_days_p50	MATCH	MATCH	MATCH	MATCH
hold_days_p99	MATCH	MATCH	MATCH	MATCH
open_hold_days	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)	EXPLAINED (final minute)
deep_share_7d	MATCH	MATCH	MATCH	MATCH
mae_vs_account_pct.worst	MATCH	MATCH	MATCH	MATCH
mae_vs_capital_pct.mean	MATCH	MATCH	MATCH	MATCH
utilization_pct.max	MATCH	MATCH	MATCH	MATCH
ladder_depth_cycles.1-5	MATCH	MATCH	MATCH	MATCH
total_volume_usd(turnover)	MATCH	n/a	n/a	n/a

Field counts: Baseline 45 exact, 4 explained, 0 unexplained; Funding 42 exact, 6 explained, 0 unexplained; Funding + 1 bp TP slip 42 exact, 6 explained, 0 unexplained; Funding + 2 bp TP slip 42 exact, 6 explained, 0 unexplained. EXPLAINED marks the two residuals in 3.2 and 3.3. The full field-level comparison (all fields, both values at full precision) is in results/scenario_comparison.csv.

Scenario	Realised wallet (ours)	Reference	Exit-net (ours, corrected)	Published exit-net	Cycles
Baseline	1,947,269.80	1,947,269.80	1,946,324.41	1,946,288.58	45,104
Funding	1,869,120.13	1,869,112.84	1,868,258.24	1,868,218.43	45,154
Funding + 1 bp TP slip	1,658,258.88	1,658,252.22	1,657,453.25	1,657,416.87	44,986
Funding + 2 bp TP slip	1,493,446.95	1,493,436.04	1,493,125.13	1,493,065.57	45,023

3.2 Residual 1 - recovered funding marks

Scenario	Realised wallet: ours - reference	Funding paid: reference - ours	Equal to 10 decimals
Funding	+7.2888003681	+7.2888003680	yes
Funding + 1 bp TP slip	+6.6610126225	+6.6610126226	yes
Funding + 2 bp TP slip	+10.9055040339	+10.9055040339	yes

Cycle counts, fees, turnover and slippage are identical in these scenarios, so the paths coincide; the only source of the wallet gap is the amount of funding paid, i.e. the recovered funding-mark inputs (3,100 of 6,186 marks were backfilled from the minute OPEN). The effect is under 0.001% of wallet. It is reported, not tuned. In the funding scenario the annual results match exactly for every year except 2026, so the recovered funding inputs differ from the originals only there.

3.3 Residual 2 - boundary inconsistency in the reference (confirmed by the client)

In all four scenarios the reference open-position hold time is exactly 189 ms longer than this report's. The last trade of the window is at 2026-08-24 23:59:59.811 UTC; adding 189 ms gives 2026-08-25 00:00:00.000. The reference runner reads mark candles up to one minute past the last trade of each day; its original mark data included the 2026-08-25 00:00 candle, so the published exit-net figures, ROI, annualised return, open-hold time and 2026 annual returns were valued at that minute's close, inside the sealed period. The client confirmed this from its code and asked for it to be reported as a boundary inconsistency. This report values at the 2026-08-24 23:59 close, the last minute of the stated window. Everything not dependent on the final minute (drawdown, liquidation room, 2026 fees, turnover, slippage, wallets) matches exactly.

Scenario	Exit-net (this report)	Published	ROI % (this report)	Annualised % (this report)
Baseline	1,946,324.41	1,946,288.58	1359.746956	60.778757

Scenario	Exit-net (this report)	Published	ROI % (this report)	Annualised % (this report)
Funding	1,868,258.24	1,868,218.43	1301.197186	59.617164
Funding + 1 bp TP slip	1,657,453.25	1,657,416.87	1143.093048	56.267836
Funding + 2 bp TP slip	1,493,125.13	1,493,065.57	1019.846650	53.404329

Corrected terminal values at full float64 precision (valued at the 2026-08-24 23:59 mark close), for the client's correction of its published figures. Also in results/terminal_values_2359.csv.

Value	Baseline	Funding	Funding + 1 bp TP slip	Funding + 2 bp TP slip
Exit-net terminal equity (USD)	1946324.4087075205	1868258.243839524	1657453.2542918387	1493125.1336104118
ROI %	1359.7469558980301	1301.1971858726076	1143.0930484515002	1019.8466498244335
Annualised ROI %	60.778756959379486	59.6171638360339	56.267835914177766	53.40432934407475
Open-position hold (days)	0.3586440625	0.3586440625	0.35864405092592594	0.3577744212962963
Open-position hold (ms)	30986847	30986847	30986846	30911710
Open position entered	2026-08-24 15:23:32.964 UTC	2026-08-24 15:23:32.964 UTC	2026-08-24 15:23:32.965 UTC	2026-08-24 15:24:48.101 UTC
Valued at (last trade)	2026-08-24 23:59:59.811 UTC	2026-08-24 23:59:59.811 UTC	2026-08-24 23:59:59.811 UTC	2026-08-24 23:59:59.811 UTC
Final mark (23:59 close)	2481.60582171	2481.60582171	2481.60582171	2481.60582171
Published exit-net (USD)	1946288.5821742606	1868218.4341446362	1657416.8734243764	1493065.5703398779
Published ROI %	1359.7200859309103	1301.1673285267984	1143.065762732689	1019.8019772598516
Published open hold (days)	0.35864625	0.35864625	0.3586462384259259	0.3577766087962963

3.4 Annual results (continuous run)

Baseline

Year	Start wallet	End wallet	Fees	Turnover	MDD %	Min liq. room %	Result
2021	133,333.00	544,154.21	83,044.51	330,080,920	14.3970	63.7817	MATCH
2022	544,154.21	903,688.48	72,764.84	289,034,145	50.2373	13.4917	MATCH
2023	903,688.48	1,038,270.11	28,027.48	108,649,678	8.9183	100.0000	MATCH
2024	1,038,270.11	1,287,049.58	51,045.54	200,361,545	34.7108	31.8324	MATCH
2025	1,287,049.58	1,703,280.67	84,774.43	334,707,317	37.0530	28.3984	MATCH
2026	1,703,280.67	1,947,269.80	49,773.33	196,332,689	17.0051	62.5071	EXPLAINED

Funding

Year	Start wallet	End wallet	Fees	Turnover	MDD %	Min liq. room %	Result
2021	133,333.00	531,044.81	81,525.48	324,140,075	14.7955	62.1420	MATCH
2022	531,044.81	881,360.11	70,836.16	281,038,107	27.1996	33.5031	MATCH
2023	881,360.11	1,002,997.11	26,443.30	102,749,289	9.2356	100.0000	MATCH
2024	1,002,997.11	1,231,320.98	48,990.28	192,090,683	36.2376	29.7536	MATCH
2025	1,231,320.98	1,635,346.29	83,651.36	329,948,741	35.0190	30.8733	MATCH
2026	1,635,346.29	1,869,120.13	47,668.34	187,994,777	17.7551	59.4773	EXPLAINED

MATCH: all thirteen annual fields equal the reference. EXPLAINED: only fields affected by the two residuals differ (funding amounts; 2026 returns valued at the final minute). The client supplied no annual reference for slip1/slip2; this report's annual results for all four scenarios are in results/annual_results.csv. Null public annual fields remain unavailable, not zero.

3.5 Rule checks and records

ID	Check	Result
E01	Initial entry: qty sized at observed price, fill = price_up(trade x (1 + initial slippage)), p1 = fill	PASS
E02	Initial taker fee on executed notional deducted from wallet immediately	PASS
E03	Ladder level 1 = price_down(p1 x (1 - cov1)); qty sized at that limit	PASS
E04	Descending crossing fills when the segment END equals the level (inclusive)	PASS
E05	No fill while price stays above the level	PASS
E06	TP at price_down(avg x (1 + TP price return)) when P0 < T <= P1; wallet += q(T - avg) - exit fee	PASS
E07	No same-trade re-entry: next cycle opens on the next trade	PASS
E08	MAE: block min INCLUDING the crossing trade, valued with the PRE-segment position (client example)	PASS
E09	Minute-start risk observation uses the minute CLOSE even without crossings	PASS
E10	Funding uses the record's own mark: payment = q x funding mark x rate	PASS
E11	Baseline (multiplier 0): no payment, but funding risk checks still run	PASS
E12	Full-ladder preview failure sets execution_invalid and stops the run	PASS
E13	Fill risk observation uses the price-fraction-interpolated mark (then the segment end at the next close)	PASS
E14	Reset base after TP only when wallet >= base x (1 + reset threshold) (no reset on a small gain)	PASS
E15	TP 749.5500000000001 (k x 0.01, one ulp above 749.55): a trade at exactly 749.55 does NOT trigger	PASS
E16	...and the next higher trade triggers, filling at the TP price	PASS

16 of 16 targeted checks pass, including the client's worked MAE example (E08) and the exact-comparison rule with the client's day-1 values (E15, E16). The client retained no cycle-level or daily-equity reference, so the reviewer-generated cycle ledgers and daily equity files (delivered privately)

cannot be compared cycle by cycle; the exact match of cycle counts, depths, holding and MAE/utilisation distributions, and of the 2021-01-01 cycle list, is the available evidence.

3.6 Phase A conclusion

Under the original assumptions the specified strategy and accounting reproduce the published ETH results bit-for-bit, apart from two explained residuals: recovered funding marks (below 0.001% of wallet) and a one-minute boundary inconsistency in the published terminal valuation, now corrected. This establishes that the published figures follow from the written rules and the data. It does not establish that those rules describe achievable exchange execution (Section 4).

4. Phase B - adequacy of the assumptions

This section assesses what the reference assumptions can and cannot support. It is ETH-specific; BTC findings were not relabelled. Nothing here altered the Phase A runs.

4.1 Marks, liquidation and risk observation

Minute-close marks applied at the minute start. Each minute's risk update and every trade in the minute use that minute's CLOSE, which is only known at the minute end. This is information from the future within each minute, and the reference observes risk only at minute starts, fills, segment ends, TPs and funding events, not at every trade or intra-minute mark. Marks inside a crossing segment are interpolated by price fraction, which is an approximation, not an observation.

Liquidation proxy: a flat maintenance rate on the whole wallet; no notional tiers, liquidation fee, partial liquidation or ADL. The closest contemporaneous distance to the proxy liquidation line was 13.4917% of mark in the baseline (funding 29.7536%); account drawdown reached 50.2373%. Zero liquidations under this proxy, observed at this resolution and with 13 proxy mark minutes in the recovered series, do not demonstrate exchange-level survival. The effect of tick-level mark observation and tiered maintenance is not quantified.

4.2 Fills, tick arithmetic and latency

A print at or through a price does not guarantee a fill. The reference fills each level in full when a trade reaches it and closes the whole position at the take-profit price, without queue, volume, partial fills or latency. It also inherits a float64 artefact: a trade exactly at the decimal TP or level price may not trigger when the computed tick price is one ulp above it (2.2); an exchange would fill at the touch. The direction of this artefact per cycle is not systematic and is not separately quantified.

Sub-second cycles: 14,897 of 45,104 baseline cycles lasted under one second (6,021 opened and closed in the same millisecond), contributing 20.2% of closed-cycle realised profit; the median cycle lasted 48.5 s. These depend on zero-latency order placement.

Rule	Definition (applied on baseline costs; every other reference rule unchanged)
F1	A level fills only if a trade prints at least one tick (0.01 USD) below it; the TP only at least one tick above it.
F2	The next level fills only once the aggTrade quantity printed at or through it since cycle entry reaches its own size; the TP only once the quantity printed at or above it since its last re-pricing reaches the position size. All-or-nothing, sequential ladder; fills use the trade's minute mark (no interpolation).
F3	As F2 with a queue ahead equal to the order's own size (2x its size must print).
L1	A new cycle opens only on the first trade at least 1 s after the TP; ladder and TP active 1 s after entry; after each additional fill the TP is re-placed and active again 1 s later.

Run (baseline costs)	Exit-net (USD)	vs reference rules	TP cycles	Liquidations	Max DD %
Reference rules (Phase A)	1,946,324.41	0.00%	45,104	0	50.24
F1 one-tick trade-through	1,970,210.55	+1.23%	45,156	0	49.90
F2 volume at/through >= 1x size	1,135,467.07	-41.66%	35,378	0	47.98
F3 volume at/through >= 2x size	1,060,026.55	-45.54%	34,198	0	51.59
L1 1-second latency	797,559.09	-59.02%	30,545	0	52.08

Limits: aggTrade quantity is a proxy for volume reaching a resting order; the true queue is unknown, so F2 and F3 bracket rather than measure queue effects; partial fills are not modelled; one latency value is tested. These are separate sensitivities, not joint execution and cost stress tests. F1 one-tick trade-through ended higher than the reference rules; stricter rules change fill prices, position sizes and cycle timing along the path, so a higher result is not evidence that the stricter execution would be favourable in live trading.

4.3 Costs, resets, drawdown and holding

The ETH baseline already includes the initial taker fee and a fixed first-entry slippage (26,874.84 USD in total), so it is not a zero-cost case. Historical funding changes the realised wallet by -4.01% versus baseline; 1 bp and 2 bp TP slippage change it by -11.28% and -20.10% versus funding only. Unlike BTC, costs change cycle counts (Baseline 45,104, Funding 45,154, Funding + 1 bp TP slip 44,986, Funding + 2 bp TP slip 45,023) because they change wallet, sizing and reset timing. The wallet reset base grew 28 times automatically; the live bot does not reinvest automatically. Longest holding 46.56 days; 20 cycles over 7 days. Fee cashback is a reporting figure only and is not in any wallet or equity figure.

5. Findings

5.1 Reproduction (Phase A). Reproduced bit-for-bit in all four scenarios, with two explained residuals (recovered funding marks; one-minute boundary inconsistency in the published terminal valuation, confirmed by the client). Corrected baseline exit-net terminal equity: 1,946,324.41 USD.

5.2 Negative findings and material limitations (Phase B).

1. Risk and liquidation use minute-close marks applied from the minute start (future information within the minute), observed only at events, with interpolated marks and a flat maintenance rate. Zero modelled liquidations do not demonstrate exchange-level survival.
2. The closest approach to the proxy liquidation line was 13.49% of mark (baseline); account drawdown reached 50.24%.
3. Fills are assumed complete when a trade reaches the price; a float64 tick artefact can delay a trigger at the exact decimal price. Queue position, partial fills and depth are not modelled.

4. Quantified execution sensitivities (baseline costs): F1 one-tick trade-through +1.23%, MDD 49.90%; F2 volume at/through $\geq 1\times$ size -41.66%, MDD 47.98%; F3 volume at/through $\geq 2\times$ size -45.54%, MDD 51.59%; L1 1-second latency -59.02%, MDD 52.08%. aggTrade volume is a proxy; the queue is assumed.

5. 14,897 cycles lasted under one second and depend on zero-latency placement.

6. Compounded results assume automatic wallet resets that the live bot does not perform.

7. The published terminal valuation used one minute of sealed-period mark data (now corrected), and funding inputs are recovered rather than original.

5.3 Unquantified effects: tick-level mark observation and mark-price liquidation timing; Binance maintenance tiers, liquidation fee and ADL; true queue position, partial fills and depth; latency other than 1 s; order rejections; live order types and fee tier; the per-cycle effect of the float64 trigger artefact; the effect of the 13 proxy mark minutes and backfilled funding marks beyond the measured funding difference.

6. Reproduction and code version

Data: <https://data.binance.vision/data/futures/um/daily/aggTrades/ETHUSDT/ETHUSDT-aggTrades-YYYY-MM-DD.zip> with matching .CHECKSUM, 2021-01-01 to 2026-08-24 (2,062 files). Run: `pip install -r requirements.txt`; `python src/downloader.py ...`; `python tests/eth_rule_checks.py`; `python run_eth.py` (one pass, all eight simulations, about one hour); `python finalize_eth.py`; `python make_packages_eth.py`. Python 3.12; numpy 2.1.3; pandas 2.2.3; numba 0.61.2; reportlab 4.2.5; matplotlib 3.9.3; requests 2.32.3; tqdm 4.67.1.

Code version v1.0.0 at <https://github.com/ranveer9/blitz-eth-tdca-verification>. The public repository contains a generic engine that requires the private configuration; without it the results cannot be reproduced.

7. Publication and attribution

Report author: Ranveer Verma. Blitz Trading may host this report in full with attribution and a profile link and may quote it accurately, subject to the agreed two-business-day technical-accuracy review of PR drafts. Blitz Trading commissioned this work; the conclusions, including the negative findings and limitations, are the author's and are preserved as found. Historical backtest results do not guarantee future or live performance.